

Immersive Block

A single EDS block for embedding interactive 3D tours, calculators, forms, quizzes, configurators, and other rich experiences directly on the page. Authors select a variant and provide a minimal identifier – all rendering logic, assets, and behavior are handled by the block. No raw HTML or manual iframe code.

The block handles three categories of immersive content under the hood, but the author simply selects the experience type from a dropdown without needing to understand how it's embedded:

- **3D Tour** – Built by the internal immersive team. Rendered natively on a canvas element using a shared viewer engine. Assets pulled from DAM.
- **Web Experience** – Built by the internal immersive team. Each experience has its own JS/CSS bundle hosted in DAM. Includes calculators, learning labs, virtual experiences, selection tools, quizzes, games, and configurators.
- **External Embed** – Built by external vendors. Embedded via iframe, constructed by the block from a provided identifier. Hosted on Thermo Fisher subdomains and third-party domains (e.g., [Kaon](#)).

Authoring Criteria

For 3D Tour:

- Experience JSON URL – required. Provided by the immersive team. The author pastes the URL in UE.

For Web Experience:

- Bundle URL – required. Provided by the immersive team. The author pastes the URL to the experience's JS bundle in UE.

For External Embed:

- Experience ID or URL is required. The author provides the identifier. The block builds the iframe internally, handling domain logic and resizing.

Authors should never paste iframe code, script tags, or raw HTML.

Hotspot popup content (3D tour) is authored as standalone chromeless pages (no header/footer), editable in UE, and loaded into modals at runtime.

Experience JSON

One per 3D tour, maintained by the immersive team. Defines everything specific to that experience:

- 3D model path and environment settings
 - Hotspot names, 3D positions, and linked content page URLs
 - Camera views and perspectives
 - Animations and trigger behavior
 - CTA buttons and links
 - Translation keys for localization
-

Config Sheet

One shared sheet for the block, maintained by the dev team. Stores infrastructure settings:

- 3D tour viewer engine JS and CSS bundle paths
- Dev and prod resource paths

Updated when the immersive team ships a new viewer version – no code deployment needed. Authors do not interact with this.

Variants

Variant	Description
3d-tour	Interactive 3D product tour with rotatable model, hotspots, side menu, camera perspectives, animations, and AR support
calculator	Interactive calculator or cost savings tool
selection-tool	Cross-reference or product selection tool
learning-lab	Interactive learning lab or learning center
virtual-experience	Virtual lab or platform experience

quiz	Quiz or assessment
game	Marketing game or gamification experience
configurator	Product configurator
form	External form experience
landing-page	Interactive landing page

Presented as a single dropdown in UE. The author selects the experience type – the block determines how to render it internally.

Development Requirements

Block code (all variants):

- Reads the selected variant and authored identifier (JSON URL, bundle URL, or experience ID/URL)
- Loads asynchronously, does not block page rendering

Block code (3D Tour):

- Creates DOM scaffolding: container div, canvas element, UI overlay mount point
- Reads viewer engine paths from the config sheet
- Loads the shared viewer engine and passes it the experience JSON URL

Viewer engine (3D Tour):

- Separate codebase, built and versioned by the immersive team, stored in the repo as vendor files
- Initializes the 3D scene from the experience JSON
- Renders hotspots, camera views, animations, side menu, and CTA buttons
- Fetches chromeless EDS pages for hotspot modals
- Platform detection for AR (iOS → .usdz, Android → .glb)

Block code (Web Experience):

- Creates a mount div for the experience
- Loads the experience's JS/CSS bundle from the authored URL
- The experience's own bundle handles all rendering and interaction

Block code (External Embed):

- Constructs the iframe from the authored experience ID/URL
- Iframe resize handling and cross-domain communication

General:

- 3D tour assets (models, textures, AR files) are hosted in DAM, not the repo
 - Native rendering (3D tour, web experience) ensures full analytics tracking
 - External iframe content loads after the EDS page – expected latency from cross-domain calls
-

Pending

External vendor content is tentatively included in this block per the discovery session. Whether these live in this block or in the embed block is TBD during implementation.